

South Asia's Climate Emergency

Floods, Heatwaves, and the Subcontinent's Civilisational Existential Risk

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Executive Summary

South Asia faces a convergence of climate threats that are simultaneously severe, rapid, and socially devastating. The IPCC identifies South Asia as a region of particular concern for heat extremes, floods, droughts, and tropical cyclones CNA[2025-11-26]. The region's climate vulnerabilities are compounded by weak infrastructure and melting glaciers — factors that climate scientists point to as amplifying the human cost of recurring extreme weather events CNA[2025-08-26]. Asia as a whole is warming nearly twice as fast as the global average, according to the WMO's State of the Climate in Asia report CNA[2025-11-26], and 2023 was confirmed as the hottest year on record by a clear margin, with records broken for ocean heat, sea level rise, Antarctic sea ice loss, and glacier retreat [RAG: CLIMATE_RAG — State of the Global Climate 2023 (WMO)].

The climate emergency is not a future projection for South Asia — it is already happening. Pakistan has experienced recurring deadly monsoon floods, with authorities in 2025 warning of multiple additional rain spells before the monsoon retreats CNA[2025-08-26]. Extreme weather events across Asia — prolonged droughts and catastrophic flooding — have highlighted the growing impact of climate change in a single season CNA[2022-08-29]. Warmer air holds more moisture, leading to heavier and more intense rainfall; the return periods for extreme events are shortening, and compound risks are increasing CNA[2025-11-26].

Part I — The Monsoon System: South Asia's Civilisational Lifeline Under Stress

The Monsoon as the Engine of South Asian Civilisation

The Indian summer monsoon is the hydraulic foundation of South Asian civilisation — the water source for hundreds of millions of smallholder farmers and the timing signal for the agricultural calendar across the subcontinent. Climate change is disrupting this system: warmer air holds more moisture, and this turns into heavier rainfall, flooding, and landslides CNA[2025-11-26]. The result is increasingly "feast or famine" monsoon behaviour — more extreme floods and more intense drought stress, sometimes in the same season.

The Monsoon Record: Increasing Extremes

Rainfall is expected to become more erratic, with shorter return periods for extreme events and higher compound risks, adding stress to countries' weather forecasting and emergency response systems CNA[2025-11-26]. Governments across the region should be investing more in climate adaptation to

minimise impacts, including upgrading canal and drainage networks and enhancing resilience to such events CNA[2025-11-26].

Pakistan has borne some of the most acute impacts. Climate scientists blame recurring floods and their high death tolls on a combination of factors: weak infrastructure and melting glaciers that amplify runoff into valleys and plains CNA[2025-08-26]. In 2025, Pakistani authorities warned of additional monsoon rain spells before the seasonal retreat, underscoring that extreme monsoon events are now a recurring rather than exceptional feature of the South Asian climate CNA[2025-08-26].

Part II — Extreme Heat: When the Planet Becomes Uninhabitable

South Asia's Heat Stress Context

The IPCC's assessment of climate risks in Asia identifies heat extremes in South Asia as one of the major concerns for the region, alongside droughts, floods, tropical cyclones, and snow cover changes [RAG: CLIMATE_RAG — IPCC AR6 Chapter 12, South/Southeast Asia regional assessment]. The WMO confirmed 2023 as the hottest year on record by a clear margin, with extreme weather events continuing to undermine socio-economic development [RAG: CLIMATE_RAG — State of the Global Climate 2023 (WMO)].

Asia is warming nearly twice as fast as the global average. This differential warming intensifies land-sea temperature gradients, amplifies pre-monsoon heat, and raises the baseline from which heat wave episodes depart CNA[2025-11-26]. Extreme weather events — from prolonged droughts to catastrophic flooding and heatwaves — have struck multiple Asian countries simultaneously, demonstrating that these are systemic rather than isolated events CNA[2022-08-29].

Part III — Flooding: When the Monsoon Becomes a Catastrophe

Pakistan: Recurring Monsoon Crisis

Pakistan has experienced repeated devastating monsoon flood seasons. Climate scientists attribute the recurring floods and high human costs to a combination of weak infrastructure and glacial melt — Himalayan and Karakoram glaciers are retreating, releasing meltwater that adds to river volumes before and during monsoon season CNA[2025-08-26]. Pakistani authorities have issued advance warnings of additional rain spells within monsoon seasons, indicating that the country now plans for extreme monsoon events as a structural condition rather than an anomaly CNA[2025-08-26].

Extreme weather events in Pakistan, documented through photographic and news records, have occurred alongside similar events in China and South Korea, confirming a regional pattern of concurrent climate extremes driven by the same underlying warming dynamic CNA[2022-08-29].

The Himalayan Cryosphere: A Deteriorating Water Tower

The Himalaya-Karakoram region is experiencing significant glacier retreat, and this retreat is creating new glacial lakes at the immediate foot of steep icy peaks with degrading permafrost and decreasing slope stability [RAG: CLIMATE_RAG — IPCC AR6 Chapter 8, Water Cycle Changes]. The risk of glacier lake outburst floods and floods from landslides into moraine-dammed lakes is increasing across Asian high mountains (high confidence) [RAG: CLIMATE_RAG — IPCC AR6 Chapter 8, Water Cycle Changes].

Mass losses of glaciers in Asia between 2000 and 2018 were -19.0 ± 2.5 Gt per year, with the most negative changes found in the Nyainqentanglha range [RAG: CLIMATE_RAG — IPCC AR6 Chapter 8, Water Cycle Changes]. While enhanced meltwater from snow and glaciers largely offsets hydrological

drought-like conditions in the short term, this effect is unsustainable: as cryospheric buffers disappear, higher temperatures in the Himalayas will lead to higher glacier melt rates, significant glacier shrinkage, and a summer runoff decrease (medium confidence) [RAG: CLIMATE_RAG — IPCC AR6 Chapter 8, Water Cycle Changes]. The long-run implication is that the river systems feeding South Asia's agricultural plains will lose their glacial buffer precisely when monsoon variability is increasing.

Part IV — Climate Finance and Adaptation

The Investment Gap

Climate finance in Asia is increasingly shifting beyond emission cuts toward adaptation and resilience, driven by growing climate shocks, energy demand, and grid pressures CNA[2026-05-20]. Bank executives speaking at Ecosperity Week 2026 noted that climate adaptation is becoming more important as environmental risks increasingly affect the businesses and projects they finance CNA[2026-05-20]. As one DBS sustainability executive stated, "with the trajectory that the global society is going from an energy transition point of view, it would be remiss for governments, societies, and banks like ourselves not to think about climate adaptation and resilience" CNA[2026-05-20].

Historically, only 16% of the \$114 billion in climate finance for developing nations over 2013 and 2014 was allocated purely for adaptation measures, with 7% more going to projects supporting both adaptation and mitigation — leaving the large majority directed at emissions reduction rather than protecting the most vulnerable populations from impacts already underway CNA[2015-10-09]. Governments in the region should be investing more in climate adaptation to minimise impacts and enhance resilience to extreme events CNA[2025-11-26].

India's Emissions Trajectory

India has a net-zero target set in policy document for 2070, placing it among G20 members whose emissions have not yet peaked [RAG: CLIMATE_RAG — UNEP Emissions Gap Report 2023-2024]. The UNEP Emissions Gap Report identifies that for countries where emissions have not yet peaked, planning to peak emissions as soon and at as low a level as feasible — while laying the foundation for decarbonization — is the relevant near-term planning focus [RAG: CLIMATE_RAG — UNEP Emissions Gap Report 2023-2024].

Part V — Air Pollution and Compounding Health Burdens

Air pollution compounds South Asia's climate vulnerability. In South and East Asia, pregnant women are frequently exposed to indoor air pollution because wood and other biomass fuels are used for cooking, which are responsible for more than 80% of regional PM2.5 pollution [RAG: HEALTH_RAG — health and air pollution compendium]. One third of preterm births globally were associated with air pollution in 2021, causing more than half a million newborn deaths [RAG: HEALTH_RAG — health and air pollution compendium]. Air pollution also increases risk of cardiovascular disease: it is responsible for 27% of deaths from strokes worldwide and 28% of coronary heart disease globally, with the risks highest in regions with higher pollution levels such as Asia [RAG: HEALTH_RAG — health and air pollution compendium].

Between 1990 and 2005 a 6% increase in global population-weighted PM2.5 was recorded, driven in particular by increased concentrations in East, South, and Southeast Asia [RAG: BIOMEDICAL_RAG — Exposure assessment for estimation of the global burden of disease attributable to outdoor air pollution (2012)]. This pollution burden intersects with climate vulnerability: heat waves reduce the effectiveness of outdoor activity that would otherwise allow families to escape indoor pollution sources, and flooding events can disrupt the clean cooking transitions needed to reduce biomass fuel dependence.

Conclusion: The Civilisational Stakes

South Asia's climate emergency is the defining challenge of this century for the region. Glacier retreat is undermining the cryospheric buffers on which hundreds of millions depend for river water [RAG: CLIMATE_RAG — IPCC AR6 Chapter 8, Water Cycle Changes]. Monsoon extremes are intensifying as Asia warms at nearly twice the global average rate CNA[2025-11-26]. Pakistan and its neighbours face recurring catastrophic flood seasons in which weak infrastructure and glacial melt combine to produce large-scale human displacement CNA[2025-08-26]. The regional air pollution burden adds a compounding health crisis that disproportionately affects the most vulnerable — pregnant women and children dependent on solid biomass fuels [RAG: HEALTH_RAG — health and air pollution compendium].

Climate finance is beginning to shift toward adaptation, but historically the balance has been heavily weighted toward mitigation — leaving adaptation chronically underfunded CNA[2015-10-09]. India's 2070 net-zero target means that the region's largest emitter will continue on an upward emissions trajectory for decades [RAG: CLIMATE_RAG — UNEP Emissions Gap Report 2023-2024], placing the burden of climate stabilisation on global action that South Asian countries cannot control unilaterally. The WMO's confirmation of 2023 as the hottest year on record — with records broken across ocean heat, sea level, ice loss, and glacier retreat simultaneously — is a signal that the pace of change is accelerating, not slowing [RAG: CLIMATE_RAG — State of the Global Climate 2023 (WMO)]. The choices made over the next decade in capitals around the world will determine which South Asia the region's people inhabit by 2100.

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